Gas Leakage Detection and Accident Prevention System

Bachelor of Technology In

ELECTRONICS AND COMMUNICATION ENGINEERING

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1. INTRODUCTION

- The project is designed to enhance home and industrial safety by automatically detecting and responding to combustible gas leaks.
- It uses the MQ-2 gas sensor, which senses gases like LPG, methane, propane, and hydrogen by detecting changes in gas concentration levels in the air.
- When the sensor detects the presence of a combustible gas above a threshold level, the system enters an emergency mode and performs several safety measures automatically.
- The 12V DC motor is activated to start ventilation, helping disperse the accumulated gas from the area.
- The LED indicator, which normally stays ON to show safe conditions, is turned OFF to visually signal a leakage condition.
- A solenoid valve connected to the gas supply is closed to prevent further gas flow, thereby reducing the risk of fire or explosion.
- The GSM module (SIM800L) sends an SMS alert to a pre-stored mobile number, warning the user about the gas leak immediately.
- If the gas concentration returns to a normal level (no leak detected), the system resumes safe conditions automatically:
    - LED turns ON to indicate safety.
    - Solenoid valve opens to restore the gas supply.
    - DC motor stops running since ventilation is no longer needed.
- The system operates efficiently with real-time monitoring and ensures quick preventive actions, reducing possible hazards from gas leaks.
- It is particularly suitable for homes, restaurants, and industries that use LPG or natural gas.

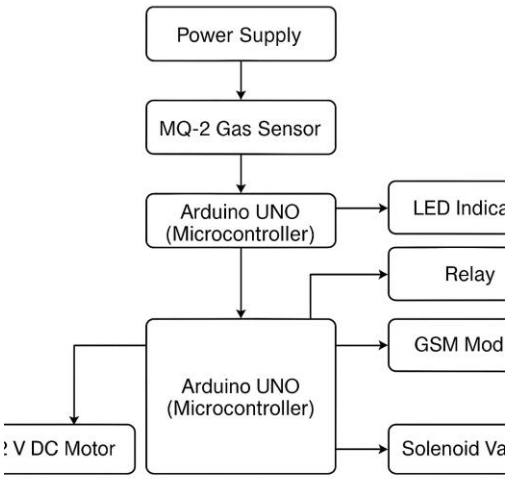
2. Relevance to Sustainable Development Goals (SDG)

This project aligns with:

- **SDG 3 – Good Health and Well-being:**
  Prevents accidents, injuries, and fatalities from gas leaks, ensuring a safer living environment.
- **SDG 9 – Industry, Innovation, and Infrastructure:**
  Promotes the use of modern sensor and microcontroller technology for innovative safety systems.
- **SDG 11 – Sustainable Cities and Communities:**
  Enhances safety and resilience in homes and industries, supporting smart and sustainable communities.
- **SDG 12 – Responsible Consumption and Production:**
  Reduces gas wastage by automatically shutting off leaks, promoting efficient resource use.
- **SDG 13 – Climate Action:**
  Minimizes greenhouse gas emissions from leaks, contributing to environmental protection.

3. Proposed System

Block Diagram:



3.1 Design Approach / Methodology:

- Identify the problem of **gas leakage** and its potential risks such as fire, explosion, and health hazards.
- Decide the need for an **automatic system** that detects gas and performs **preventive and alerting actions**.
- Design the **block diagram** showing all major components — MQ-2 sensor, Arduino UNO, relay module, GSM module, motor, solenoid valve, and LED.
- Determine **power requirements** (5V for control circuit and 12V for output devices) and ensure a **common ground connection**.
- Interface the **MQ-2 gas sensor** with Arduino through the analog pin (A0) to sense gas concentration.
- **Calibrate** the sensor by testing it in clean air and near gas leaks to set a suitable **threshold value**.
- Program the **Arduino UNO** to continuously read the analog value from the MQ-2 sensor.
- Compare the sensor value with the **threshold** to decide whether a gas leak is detected or not.
- When gas is detected:
  - Turn **ON the motor** to ventilate the area.
  - **Close the solenoid valve** to stop gas flow.
  - Turn **OFF the LED** to indicate danger.
  - **Send SMS alert** to the user via the GSM module.
- When no gas is detected:
  - Keep **LED ON** (normal condition).
  - Keep **motor OFF** and **solenoid valve open**.
- Use **relay modules** to safely control high-current devices (motor and solenoid) from Arduino.
- Interface the **GSM module (SIM800L)** with Arduino using TX and RX pins for sending SMS notifications.
- Include a **time delay** in code to prevent repeated or spam messages.
- Test the system with varying gas levels to verify accuracy and reliability.
- Finalize the **hardware connections** on a breadboard or PCB after successful testing.
- Ensure the system runs automatically with the workflow: **Sense → Process → Actuate → Alert**.

3.2 Code:

```
#include <SoftwareSerial.h>


// Pin connections

int mqSensor = A0;           // MQ-2 sensor connected to A0

int motorRelay = 6;          // Relay to control motor

int solenoidRelay = 7;       // Relay to control solenoid valve

int ledPin = 8;              // LED for status indication


// GSM module pins

int gsmRx = 10;              // Arduino RX (connected to GSM TX)

int gsmTx = 11;              // Arduino TX (connected to GSM RX)

SoftwareSerial gsm(gsmRx, gsmTx);  // RX, TX


// Variables

int gasValue = 0;            // Stores gas sensor value

int threshold = 350;         // Set limit after testing

unsigned long lastSmsTime = 0;

unsigned long smsDelay = 120000; // 2 minutes gap between
messages


// Change to your mobile number

char phoneNumber[] = "+91XXXXXXXXXX";


void setup() {

  Serial.begin(115200);

  gsm.begin(9600);


  pinMode(mqSensor, INPUT);
```

```
  pinMode(motorRelay, OUTPUT);
  pinMode(solenoidRelay, OUTPUT);
  pinMode(ledPin, OUTPUT);

  // Default safe state
  digitalWrite(motorRelay, HIGH);      // Motor OFF (Relay
normally closed)
  digitalWrite(solenoidRelay, HIGH);   // Solenoid Open
  digitalWrite(ledPin, HIGH);          // LED ON (System normal)

  Serial.println("Gas Leakage Detection System Starting...");
  delay(2000);

  // GSM initialization
  sendAT("AT");
  delay(500);
  sendAT("AT+CMGF=1"); // Set to text mode
  delay(500);
}

void loop() {
  gasValue = analogRead(mqSensor); // Read gas sensor
  Serial.print("Gas Sensor Value: ");
  Serial.println(gasValue);

  if (gasValue > threshold) {
    gasDetected();     // Run alarm functions
  } else {
    systemNormal();    // Stay normal
```

```
  }

  delay(1000); // Check every second
}

void gasDetected() {
  Serial.println("ALERT! Gas Leak Detected!");

  // Turn ON safety devices
  digitalWrite(motorRelay, LOW);       // Motor ON
  digitalWrite(solenoidRelay, LOW);    // Solenoid Closed
  digitalWrite(ledPin, LOW);           // LED OFF

  // Send SMS alert (only once every few minutes)
  if (millis() - lastSmsTime > smsDelay) {
    sendSMS(phoneNumber, "ALERT: Gas Leak Detected! Please Check
Immediately!");
    lastSmsTime = millis();
  }
}

void systemNormal() {
  Serial.println("System Normal - No Gas Detected.");

  // Keep devices OFF / Open
  digitalWrite(motorRelay, HIGH);     // Motor OFF
  digitalWrite(solenoidRelay, HIGH);  // Solenoid Open
  digitalWrite(ledPin, HIGH);         // LED ON
}
```

```
void sendAT(const char *cmd) {
  gsm.println(cmd);
  delay(500);
  while (gsm.available()) {
    Serial.write(gsm.read());
  }
}

void sendSMS(const char *number, const char *msg) {
  Serial.println("Sending SMS...");
  gsm.println("AT+CMGF=1"); // Text mode
  delay(300);
  gsm.print("AT+CMGS=\"");
  gsm.print(number);
  gsm.println("\"");
  delay(300);
  gsm.print(msg);
  delay(300);
  gsm.write(26); // CTRL+Z to send SMS
  delay(3000);
  Serial.println("SMS Sent Successfully!");
}
```

4. Hardware and Software Tools used

1. Hardware:

- **Arduino UNO** – Acts as the main microcontroller that processes the data from the gas sensor and controls the outputs like motor, LED, solenoid valve, and GSM module.

- **MQ-2 Gas Sensor** – Detects the presence and concentration of combustible gases such as LPG, methane, propane, and smoke.

- **GSM Module (SIM800L / SIM900)** – Sends SMS alerts to the user's mobile number when gas leakage is detected.

- **Relay Module (5V Single Channel)** – Electrically isolates and switches high-power devices such as the motor and solenoid valve.

- **12V DC Motor** – Works as an exhaust fan to remove leaked gas from the area.

- **12V Solenoid Valve** – Automatically closes the gas supply line to prevent further leakage.

- **LED Indicator** – Provides visual indication of system status (ON = safe, OFF = leak).

- **Power Supply (12V and 5V)** – Provides regulated DC power to the Arduino, sensors, GSM module, and output devices.

- **Connecting Wires and Breadboard / PCB** – Used for circuit connections and component placement.

2. Software:

- **Arduino IDE –** Used for writing, compiling, and uploading the Arduino program to the microcontroller.

5. Conclusion

- This project **"Gas Leakage Detection and Prevention System"** successfully senses the presence of combustible gases such as LPG, methane, and propane using the **MQ-2 sensor**.

- The system automatically operates to **prevent accidents** by activating safety devices and **alerting users in real time**.

- When a gas leak is detected, the **Arduino UNO** processes the sensor signal and immediately turns **ON the exhaust motor**, **closes the solenoid valve**, **turns OFF the LED**, and **sends an SMS alert** through the **GSM module**.

- Under normal conditions, the system keeps the **LED ON**, **motor OFF**, and **solenoid open**, indicating a safe environment.

- The system is **low-cost, efficient, and easy to implement**, making it suitable for **homes, industries, hotels, and laboratories** where gas safety is critical.

- The integration of **analog sensing, digital control, and wireless communication** demonstrates the effective use of microcontroller-based automation for real-time safety applications.

- Overall, the project provides a **reliable and practical solution** for preventing gas-related accidents and protecting human life and property.